

Mathematics

Stage 8

Paper 2

2025

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Calculator
Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Expand the brackets.

$$7m(5m + 2)$$

..... [1]

- 2 Pierre works in a factory employing 400 people.
He gives this question to a sample of 5 people in the factory.

Draw a ring around the score that best shows how you feel about your job.									
1	2	3	4	5	6	7	8	9	10
unhappy									very happy

The mean score for the 5 people is 2.6

Pierre's conclusion is that **all** the people in the factory are unhappy with their job.

Explain why Pierre's conclusion may **not** be correct.

..... [1]
.....

- 3 Complete the table by rounding each number correct to 2 significant figures.

Number	Number rounded correct to 2 significant figures
6324	
0.7286	

[2]

4 Complete this sentence.

45 miles is equal to km.

[1]

5 (a) Draw a ring around the inequality that is **not** equivalent to $x < 16$

$3x < 48$

$x + 8 < 22$

$x - 18 < -2$

$\frac{x}{8} < 2$

[1]

(b) In this question $25 < y < 35$

Write down the smallest integer value of y .

..... [1]

6 A 3D shape has 5 vertices and 5 faces.

Work out the number of edges on this shape.

..... [1]

- 7 Yuri thinks of a whole number greater than 0
He writes some statements about multiplying and dividing his number by powers of 10

Tick (✓) to show if each statement is true or false.

	True	False
Dividing my number by 10^3 will make my number smaller.	☐	☐
Multiplying my number by 10^1 will not change my number.	☐	☐
Dividing my number by 10^2 will make my number 20 times smaller.	☐	☐
Multiplying my number by 10^4 will make my number 10 000 times bigger.	☐	☐

[2]

- 8 Draw a line to match each formula, equation or expression to the correct description.

$x - 3 = 20$	Formula
$10x + 2$	Equation
$A = \frac{1}{2} \times b \times h$	Expression

[1]

- 9 A number rounded correct to 2 decimal places is 3.87

Draw a ring around **each** of the values this number could be.

3.875

3.8749

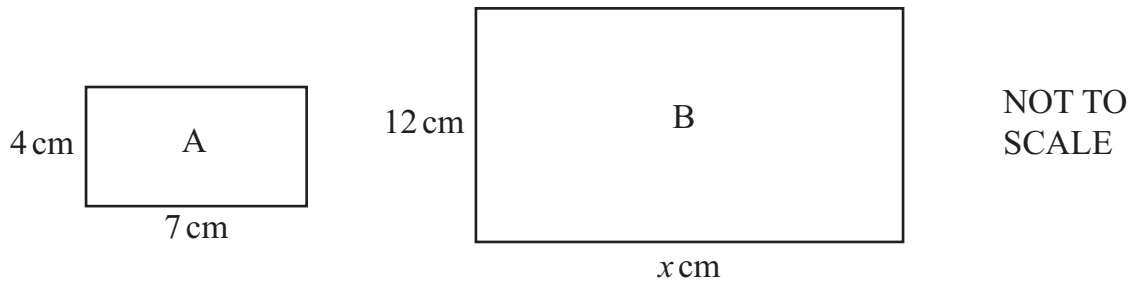
3.865

3.8645

3.8649

[1]

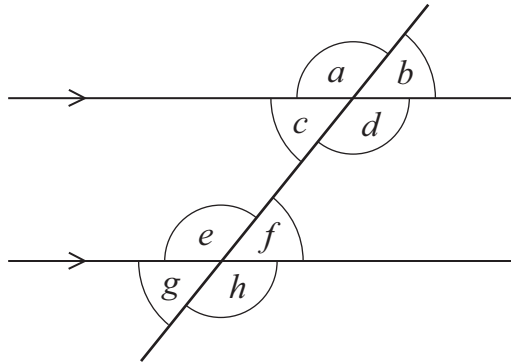
- 10 The diagram shows two rectangles, A and B.
Rectangle A is enlarged to give rectangle B.



Work out the value of x .

$x =$ [1]

- 11** The diagram shows a pair of parallel lines with a line crossing them. The angles are marked with letters.



Complete these sentences.

The angle that is vertically opposite to angle b is

The angle that is corresponding to angle e is

The angle that is alternate to angle c is

[2]

- 12** Factorise.

$$15x - 25$$

.....

$$26x^2 + 39x$$

.....

[3]

13 Tick (✓) the **two** lines that are parallel to each other.

$$y = 3x - 7$$

☐

$$y = 7x - 3$$

☐

$$y = 7 - 3x$$

☐

$$y = 7 + 3x$$

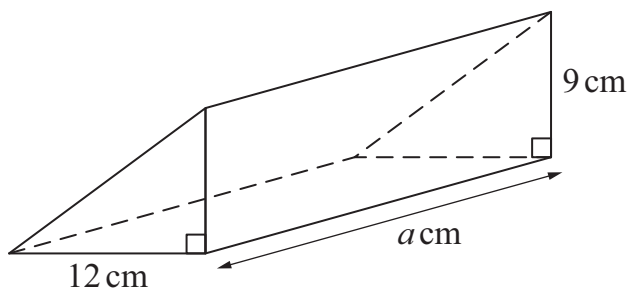
☐

$$y = 3 - 7x$$

☐

[1]

14 The diagram shows a prism.



NOT TO
SCALE

The volume of the prism is 1269 cm^3 .
The length of the prism is $a \text{ cm}$.

Calculate the value of a .

$a = \dots\dots\dots$ [2]

15 Hassan has a box of chocolates.

$\frac{1}{5}$ of the chocolates are milk chocolates.

$\frac{3}{10}$ of the chocolates are white chocolates.

The rest of the chocolates are dark chocolates.

Write down the ratio of **milk : white : dark** chocolates.
Give the ratio in its simplest form.

..... : : [3]

16 Angelique wants to investigate the favourite food of students in her school.

(a) Angelique is deciding how to collect her data.

Complete each of Angelique's sentences with the most appropriate word from the box.
You may use each word more than once.

questionnaires	interviews
----------------	------------

Using will take me a long time as I need to speak to everyone.

Using means the questions can be explained if students do not understand.

Using means I can collect data from a large sample of students all at the same time.

[1]

(b) There are 500 girls and 550 boys in Angelique's school.
Angelique decides to sample 50 girls and 55 boys from her school.

Explain why this is an appropriate sample.

.....
..... [1]

(c) Angelique is trying to decide how to pick the 50 girls for her sample.
She decides to pick the first 50 girls she sees coming into her school.

Write down one advantage of using this method.

.....
..... [1]

17 Here is sequence A.

2, 5, 8, 11, ...

Here are the rules to make sequences B, C and D.

Double each term in sequence A to make sequence B.

Add 2 to each term in sequence B to make sequence C.

Divide each term in sequence C by 3 to make sequence D.

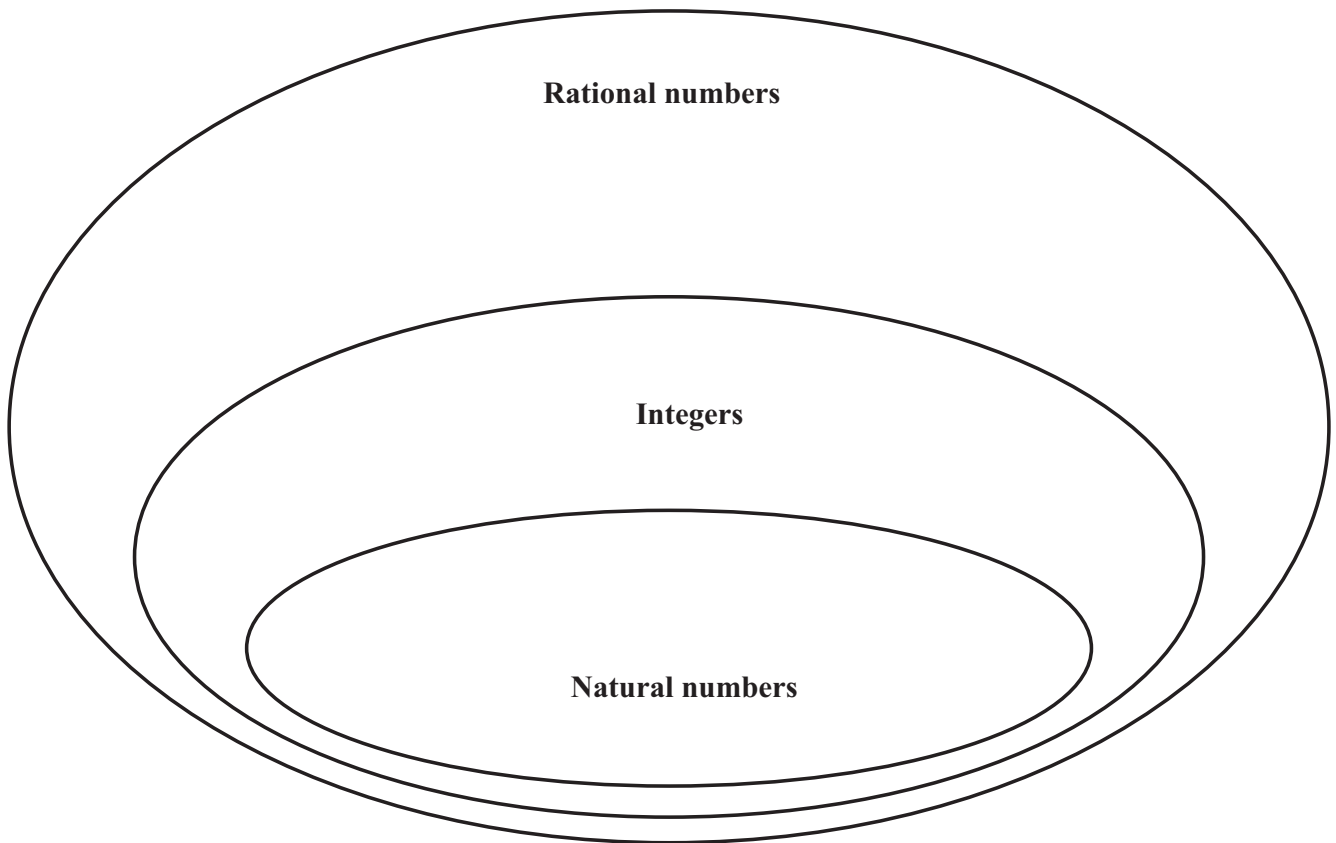
Complete these sentences.

The term-to-term rule for sequence D is

The term-to-term rules are the same for sequence and sequence

[2]

18 Here is a Venn diagram.



Write each of these numbers in the correct place on the Venn diagram.

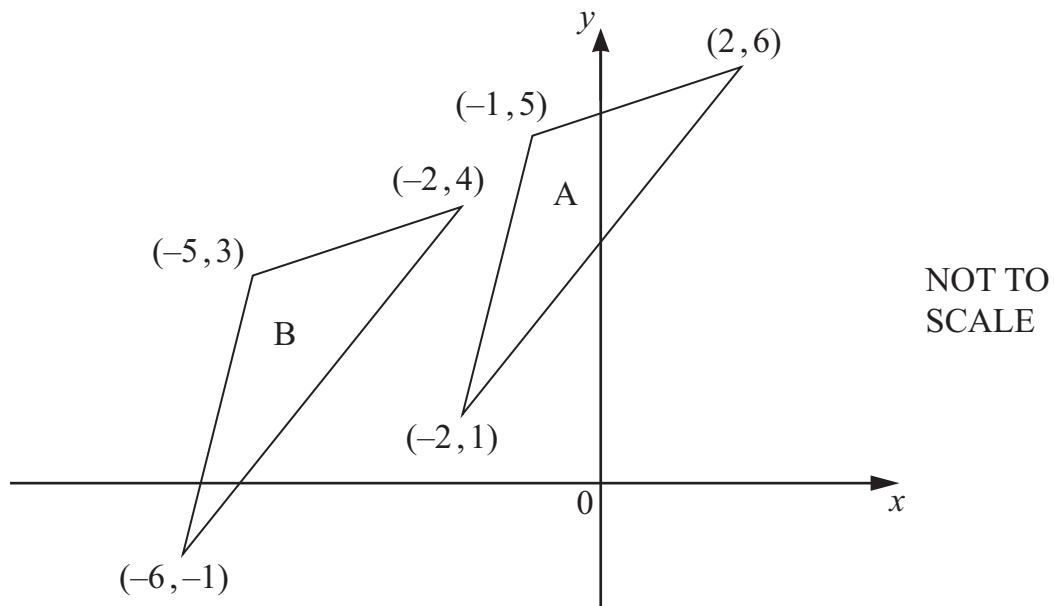
-5 0.15 6 153.2 $\frac{2}{3}$ -1

[2]

19 Find the value of $10x^2$ when $x = -2$

..... [1]

20 Here are two triangles, A and B, with the coordinates of each vertex shown.



Find the vector that translates triangle A onto triangle B.

$\begin{pmatrix} \\ \end{pmatrix}$ [1]

21 Here are two numbers written as a product of their prime factors.

$$2^5 \times 3 \times 5^2 \times 7$$

$$2^3 \times 3^2 \times 5^3$$

h is the highest common factor of the two numbers.

l is the lowest common multiple of the two numbers.

Find the value of h and the value of l .

$h =$

$l =$

[3]

- 22** Each of the adults in group A and group B runs a race.
 The time, in seconds, to run the race is recorded for each adult.
 The stem-and-leaf diagram shows the results for each adult in group A.

Group A

6	2	4	6		
7	0	2	2	7	9
8	1	1	1	7	
9	3	4	8		

Key: **9** | 3 represents a time of 93 seconds to run the race.

- (a)** Write down the fastest time for the adults in group A to run the race.

..... seconds [1]

- (b)** Find how many adults in group A took longer than 80 seconds to run the race.

..... [1]

- (c)** Ahmed compares the range of times for group A with the range of times for group B.
 The ranges show that group A has more consistent times than group B.

Draw a ring around the value that could be the range for **group B**.

30 32 34 36 38

[1]

- (d)** The median time for group B adults to run the race is 82 seconds.

Using this result, write down a comparison between group A and group B.
 You must use data to support your comparison.

.....
 [2]

- 23 Here are the first four terms of a linear sequence.

3.4 2.8 2.2 1.6

Find the n th term of this sequence.

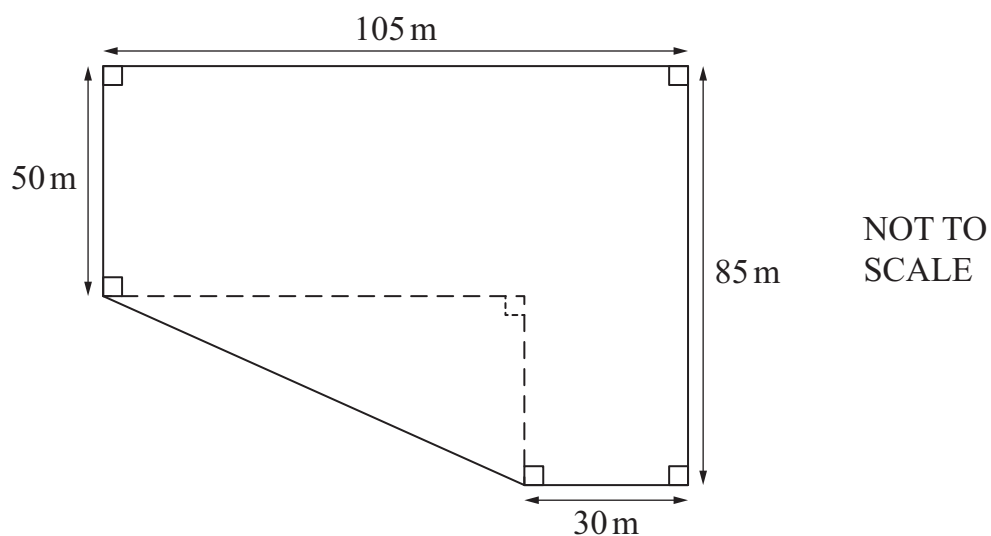
..... [2]

- 24 The wheel of a car has a radius of 22.5 cm.
The car travels 1.1 **km**.

Work out the number of times the wheel turns when the car travels 1.1 **km**.
Give your answer correct to the nearest whole number.

..... [3]

25 The diagram shows a field in the shape of two rectangles and a triangle.



Mia wants to cover the field with grass seed.

Grass seed is sold in 25 kg bags.

5 kg of grass seed covers 115 m^2 .

Mia thinks she will need to buy 13 bags of grass seed.

Show that Mia is **not** correct.

[4]

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